

Multi-Seed Reproducibility and Convergence Stability of Non-IID Federated YOLO11n for Oil Palm Fresh Fruit Bunch Detection

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Abstract

Federated learning (FL) enables collaborative model training across distributed data holders without centralizing raw images, which is attractive for oil palm plantation operators facing data-sharing and bandwidth constraints. However, FL deployments in agricultural computer vision are frequently reported from a single training run, leaving open how reproducible their reported performance is under independent stochastic training realizations. This study evaluates the reproducibility and convergence stability of Federated Averaging (FedAvg) for six-class oil palm fresh fruit bunch (FFB) ripeness detection using YOLO11n with Group Normalization, under a fixed Non-IID client partition (Dirichlet $\alpha = 0.5$, $K=4$ simulated clients, training-image share ranging 8.67%–59.36% per client) on a leakage-free, source-group-aware dataset split (8,937 training / 826 validation / 1,051 held-out test images). Training was replicated across three independent seeds (42, 123, 2026) over 40 communication rounds each. Validation mAP50 converged to a stable mean of 0.8775 (SD = 0.0157) across seeds, but the best-validation checkpoint occurred at markedly different rounds (9, 11, and 40, respectively), indicating that outcome-level reproducibility does not imply trajectory-level reproducibility. A centralized reference model, evaluated once on the held-out test split, achieved mAP50 = 0.8820 (mAP50-95 = 0.7309), which is reported descriptively alongside – not as a matched comparison against – the federated validation results. These findings suggest that multi-seed reporting is necessary to characterize federated training behavior under Non-IID partitions, and that checkpoint-selection policy should account for seed-dependent convergence speed. A matched held-out-test evaluation of the federated checkpoints is identified as the immediate next step.

Keywords: Federated Learning; Non-IID Data; Reproducibility; Convergence Stability; Oil Palm Ripeness Detection; YOLO11.

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INTRODUCTION

Precise, non-destructive assessment of fresh fruit bunch (FFB) ripeness is a long-standing operational problem in oil palm cultivation, since harvesting decisions made from ripeness stage directly affect oil extraction rate and downstream yield [4, 7]. Because ripeness assessment in commercial operations is still predominantly performed through manual visual inspection, harvesting decisions can be inconsistent across harvesters and across observation conditions such as lighting and viewing angle, which in turn motivates automated, image-based ripeness assessment as a means of supporting more objective and repeatable harvesting decisions [4, 7]. Deep-learning-based object detectors, and in particular the YOLO family, have been widely adopted for this task because they combine real-time inference speed

with competitive detection accuracy [1, 2]. In practical deployments, FFB image data are frequently distributed across multiple sources (e.g., different harvesting sites, collection batches, or partner organizations) rather than residing in a single centralized repository. Centralizing such data for model training raises data-sharing and logistical concerns, which motivates the use of Federated Learning (FL) [12, 13] as a means of collaborative model training without transmitting raw images to a central server.

FL allows multiple clients to collaboratively train a shared model by exchanging model updates rather than raw data [12]. It is important to note that FL, by itself, does not constitute a formal privacy guarantee: model updates and gradients can still leak information about the underlying training samples [17, 18]. Formal guarantees require mechanisms such as Differential Privacy (DP-SGD), which are outside the scope of the present study and are noted only briefly as future work in the Discussion section. The property this study relies on is narrower and purely logistical: FL avoids the specific operational requirement of centralizing raw plantation imagery on a single server.

A separate, and for deployment purposes equally important, question is *reproducibility*. Because FedAvg optimization under Non-IID client data is a stochastic process influenced by client sampling order, local minibatch composition, and weight-initialization-adjacent randomness, a single federated training run cannot by itself establish whether a reported performance level is a stable, repeatable outcome or an artifact of one favorable stochastic trajectory. This distinction matters operationally: a practitioner deciding whether to deploy a federated-trained model needs to know not only its peak achievable performance, but whether that performance – and the number of communication rounds required to reach it – is consistent across independent training runs.

Research Gap

Prior work has separately examined YOLO-based object detection for oil palm FFB ripeness [3, 4, 5, 6, 7] and Federated Learning for visual classification and detection under Non-IID data [12, 14, 16]. However, federated learning studies in agricultural computer vision typically report results from a single training run, without characterizing performance variance or convergence-trajectory variance across independent stochastic seeds. This gap is consequential because single-run reports cannot distinguish a reproducible training regime from a favorable outlier run, which is precisely the distinction this study is designed to test. This study does not claim to be the first application of Federated Learning to oil palm imagery or the first use of YOLO for FFB ripeness classification; rather, it establishes a multi-seed empirical characterization of reproducibility and convergence behavior for this specific task, architecture, and federation setting.

Research Question

The primary research question addressed in this study is: under a fixed Non-IID client partition (Dirichlet $\alpha = 0.5$, $K=4$), how reproducible is FedAvg-trained YOLO11n's detection performance across independent stochastic training seeds, and how consistent is its convergence behavior across communication rounds?

This is examined through three supporting questions: (i) does peak validation performance remain stable across training seeds even when the communication round at which that peak occurs varies substantially; (ii) does severe client data-quantity imbalance (8.67%–59.36% of training images per client) destabilize FedAvg convergence under sample-weighted aggregation; and (iii) is federated training's per-class detection stability consistent with its global mAP stability, or do specific ripeness classes show disproportionate inter-seed variance that a single aggregate metric would hide.

Contributions

(1) A multi-seed ($n = 3$) empirical evaluation of FedAvg under a fixed Non-IID Dirichlet partition ($\alpha = 0.5$, $K=4$) for six-class oil palm FFB ripeness detection using YOLO11n, reporting outcome-level reproducibility (mean validation mAP50 = 0.8775, SD = 0.0157). (2) A convergence-trajectory analysis across 40 communication rounds per seed, showing that the best-validation checkpoint round varies substantially across seeds (round 9, 11, and 40) despite convergent final performance – distinguishing outcome reproducibility from trajectory reproducibility. (3) A quantitative characterization of federated client heterogeneity by training-image count (8.67%–59.36% share across four simulated clients under Dirichlet partitioning; coefficient of variation ≈ 0.84), motivating discussion of sample-weighted FedAvg aggregation dynamics. (4) A leakage-audited, source-group-aware train/validation/held-out-test protocol that separates the reproducibility analysis (validation split, multi-seed) from a single locked centralized reference point (held-out test split). (5) An architectural adaptation (Batch Normalization to Group Normalization throughout) applied consistently across all training configurations, reported here as a design choice compatible with small Non-IID per-client batches, not as an empirically demonstrated performance improvement. Centralized training (B1) is retained strictly as a single reference point, not as a co-equal contribution.

Related Work

Oil palm FFB detection and grading. Deep-learning-based detection of oil palm FFB ripeness has predominantly used one-stage object detectors, particularly YOLO-based architectures built on convolutional backbones such as CSPNet [9] and residual networks [8], for their real-time performance [3, 4, 5, 6, 7]. Lai et al. applied YOLOv4 to real-time ripe-bunch detection in plantation settings [3], while Suharjito et al. targeted mobile-device deployment for FFB ripeness classification [4]; a recent review notes that most reported systems remain single-site, centrally trained models [7].

Federated learning for computer vision and object detection. FedAvg [12] remains the standard aggregation baseline for federated visual learning. Federated learning under Non-IID client data has been studied for general vision classification tasks [14, 16], but agricultural-domain federated object detection specifically remains comparatively under-reported [TODO: NEEDS RECENT REFERENCE: a 2023–2025 paper applying FedAvg or a federated variant to an agricultural detection/classification task with multiple clients].

Non-IID heterogeneity and convergence stability. Non-IID client data is a well-documented source of federated optimization instability [16, 14]. Normalization schemes that avoid dependence on cross-sample batch statistics, such as Group Normalization [11], have been proposed as one mitigation for batch-statistic instability under small, heterogeneous per-client batches [15]. [TODO: NEEDS RECENT REFERENCE: a 2022–2025 paper specifically analyzing convergence-round variance (not only final-accuracy variance) under Dirichlet-partitioned Non-IID FL – the closest prior work to this paper’s core claim; suggested search terms: "federated learning convergence round variance", "FedAvg training seed variance Non-IID", "checkpoint selection federated learning heterogeneous clients".]

Reproducibility and multi-seed evaluation. Multi-seed reporting is an established methodology for characterizing training variance in general deep learning [TODO: NEEDS RECENT REFERENCE: a general ML reproducibility paper on multi-seed reporting for benchmark claims; search terms: "reproducibility deep learning benchmarks seed variance"], but its adoption in federated learning research specifically – as opposed to centralized training – is comparatively rare, particularly for object detection tasks in applied domains such as agriculture [TODO: NEEDS RECENT REFERENCE: a paper

specifically calling for or performing multi-seed federated learning evaluation; search terms: "multi-seed federated learning evaluation", "federated learning experimental reproducibility".] This study addresses this combined gap by reporting three independent training seeds under a fixed Non-IID partition for a federated object-detection task, characterizing both outcome-level and trajectory-level reproducibility – a distinction not, to the authors' knowledge, jointly reported for agricultural federated object detection.

MATERIALS AND METHODS

Figure 1 summarizes the overall research workflow described in this section, from dataset preparation through the centralized (B1) and Non-IID federated (B2) training and evaluation paths.

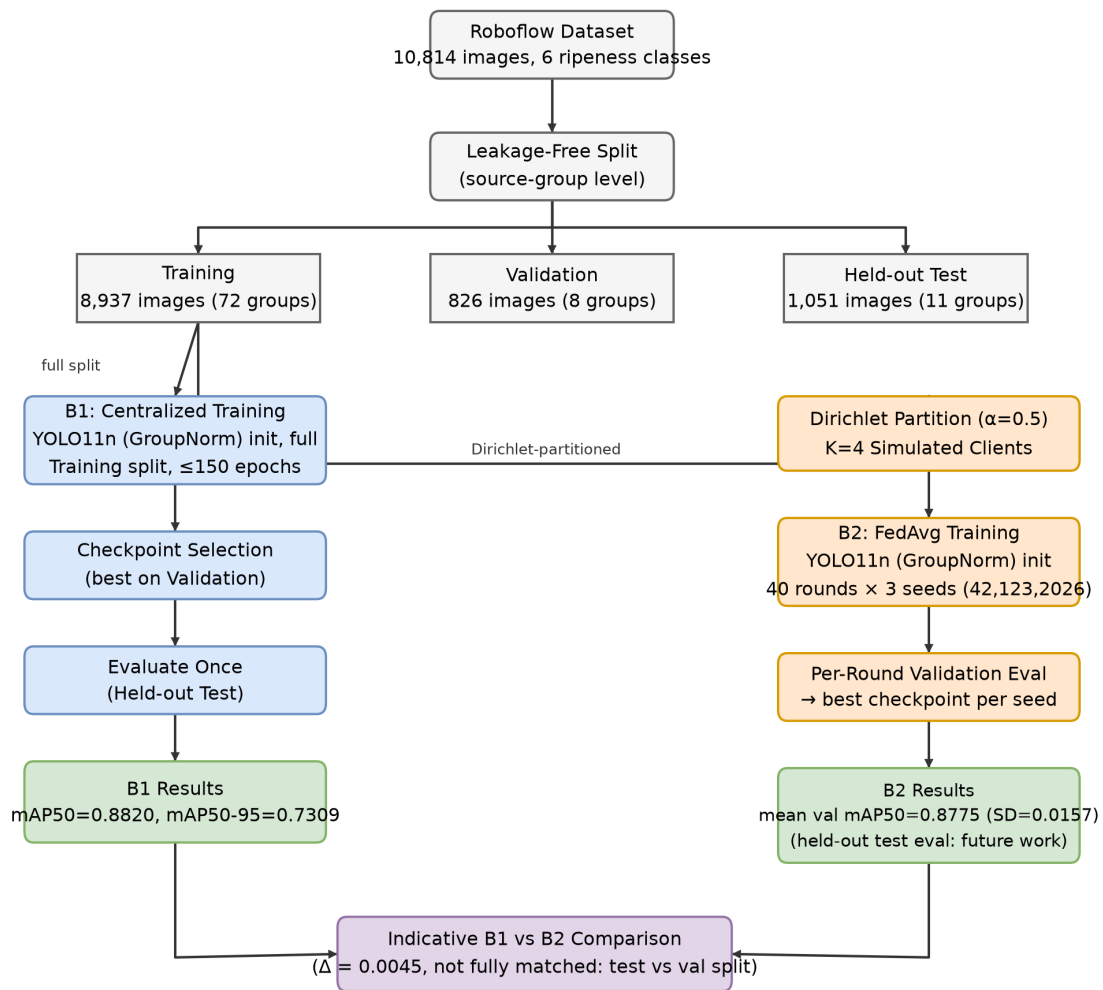


Figure 1: Overall research workflow, from dataset preparation through the centralized (B1) and Non-IID federated (B2) training and evaluation paths.

Dataset

This study uses the Palm Fruit Ripeness Detection dataset (version 2) sourced from Roboflow, comprising 10,814 images annotated for six ripeness-related classes: Abnormal, Empty Bunch, Overripe, Ripe, Underripe, and Unripe, in a manner broadly consistent with other publicly documented efforts to build annotated FFB ripeness datasets for deep-learning-based grading [5]. The six classes reflect

an operational grading protocol rather than a strict physiological maturity scale. Unripe and Underripe capture fruit at progressively earlier stages of colour change with little to no loose fruit; Ripe and Overripe correspond to the harvest-optimal and post-optimal stages, distinguished mainly by the proportion of loose fruit and the extent of colour development; and Empty Bunch and Abnormal represent non-standard bunch conditions, respectively bunches with most fruit already detached and bunches whose shape, density, or development departs from the normal pattern, rather than points on the ripeness continuum. This class structure is relevant to interpreting the per-class results reported in *B1: Centralized Reference Baseline Results* below, since Ripe and Overripe are visually contiguous with their neighboring stages, whereas Empty Bunch and Abnormal are visually more distinct outlier conditions. Figure 2 shows a representative example image for each of the six classes.

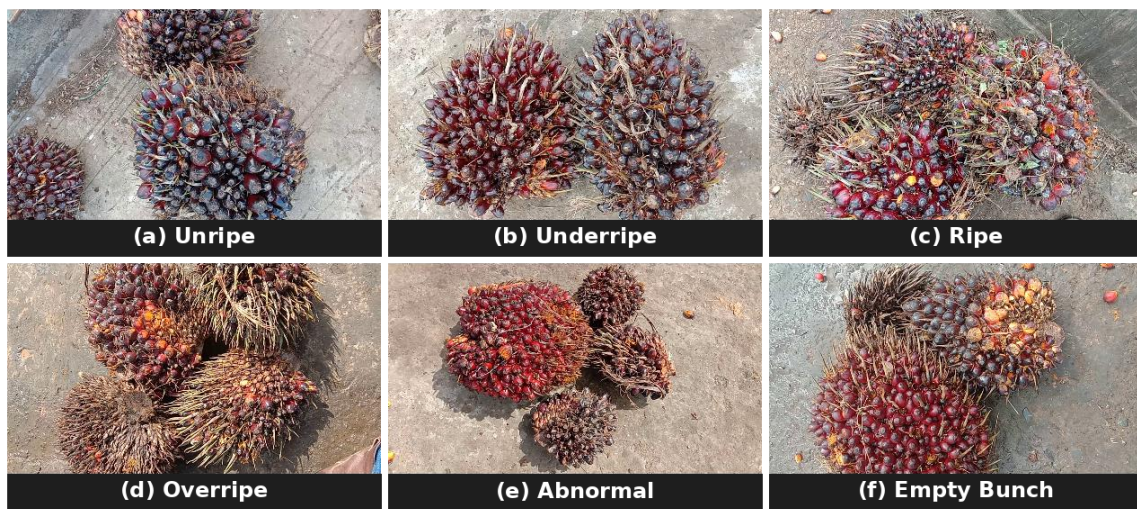


Figure 2: Representative example images for each of the six ripeness-related classes (Abnormal, Empty Bunch, Overripe, Ripe, Underripe, Unripe).

Leakage-Free Split

Images are grouped by their originating source group prior to splitting, and the train/validation/held-out-test partition is constructed at the source-group level to prevent source-related leakage across splits. The resulting split is summarized in Table 1. A leakage audit was performed to confirm: no source-group overlap between the training, validation, and test splits; no identical cross-split duplicate images; and no confirmed cross-split duplicate overlap according to the implemented audit procedure.

Table 1: Leakage-free dataset split

Split	Images	Source groups
Training	8,937	72
Validation	826	8
Held-out test	1,051	11

Federated Partition

Only the training subset (8,937 images) is partitioned among federated clients; the validation and held-out test splits are not partitioned. The federation uses $K=4$ clients under a Non-IID partition generated with a

Dirichlet distribution ($\alpha = 0.5$, partition seed = 42). The Dirichlet distribution is a standard mechanism for constructing controllable Non-IID splits in federated learning research, since its concentration parameter directly governs the degree of class-distribution skew across clients without requiring additional heuristic allocation rules [16]. Table 2 and Figure 3 report the resulting client-level image distribution. The clients are simulated sample-level data shards and do not represent four physical plantations, estates, organizations, or devices; training is simulated sequentially on a single GPU. Images originating from one source group inside the training subset may be distributed across different clients; this does not violate the leakage-free train/validation/test split, since all client partitions are drawn exclusively from the training subset.

Table 2: Federated client image distribution (K=4, Dirichlet $\alpha = 0.5$, seed=42)

Client	Images	Share (%)
Client 0	860	9.62
Client 1	775	8.67
Client 2	5,305	59.36
Client 3	1,997	22.35
Total	8,937	100.00

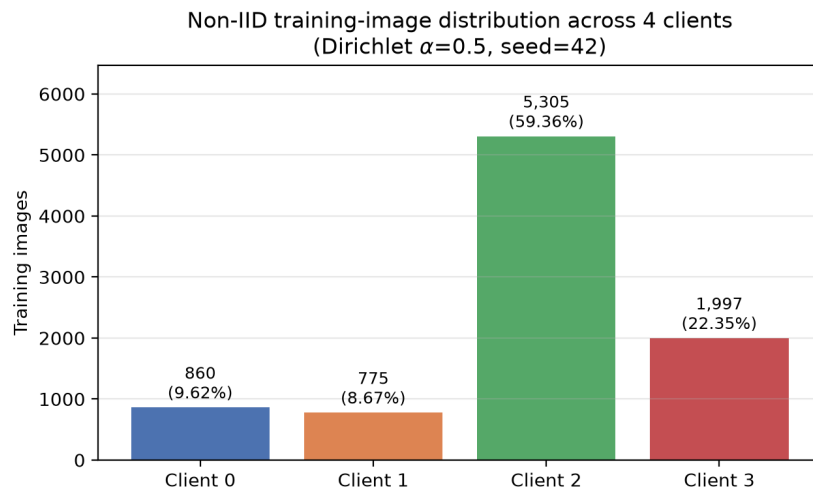


Figure 3: Non-IID training-image distribution across the four simulated federated clients (Dirichlet $\alpha = 0.5$, seed 42).

Client Heterogeneity Measurement

Table 2 quantifies client heterogeneity by training-image count. As a scalar summary of this quantity imbalance, the coefficient of variation across the four clients' image counts (860, 775, 5,305, 1,997) is $CV = SD/\text{mean} \approx 0.836$, indicating substantial dispersion relative to the mean client size. This measurement establishes quantity-level heterogeneity only; it does not by itself establish class-distribution skew across clients, since a client could hold few images while still preserving the global class proportions. [TODO: AUTHOR INPUT REQUIRED: a per-client, per-class instance count table (from the partition manifest produced by scripts/03_partition_clients.py) would additionally quantify class-level

heterogeneity, e.g. via per-client normalized entropy or Jensen–Shannon divergence from the global class distribution; this is deferred to future reporting and is not required for the reproducibility analysis in this manuscript, which does not depend on class-level skew.]

Key Terminology

To avoid ambiguity in the multi-seed protocol described below, this study distinguishes: the *partition seed*, which controls the Dirichlet client-partition assignment and is fixed at 42 for all runs so that all three training seeds share an identical client data allocation; the *training seed*, which controls stochastic elements of the training process itself (data shuffling order and augmentation sampling) and is varied across 42, 123, and 2026; a *communication round*, one complete cycle of local training at all four clients followed by FedAvg aggregation (40 total per seed); a *local epoch*, one full pass over a single client’s local training partition within one communication round (2 per round); the *best-validation checkpoint*, the model state saved after the communication round with the highest validation mAP50 for that seed; and the *final-round checkpoint*, the model state after the last (40th) communication round, regardless of whether it is also the best-validation checkpoint.

Model Architecture

The base detector is YOLO11n [2], initialized from weights pretrained on the COCO dataset [10], with an input resolution of 960×960 . The adjusted model contains 2,591,010 parameters. All Batch Normalization layers were converted to Group Normalization [11], since cross-sample batch statistics are known to become unreliable under small, Non-IID per-client batches in federated settings [15]; a final architecture audit confirmed 0 remaining BatchNorm modules and 81 GroupNorm modules across the network. This substitution is applied consistently across B1 and B2 so that normalization type is not a confounding factor when interpreting results, and is reported here strictly as a design choice, not as an empirically demonstrated performance or stability improvement, since no BatchNorm-versus-GroupNorm ablation was run in this study (see *Limitations*). A fixed convolution inside the Distribution Focal Loss (DFL) module is set with `requires_grad=False` and is excluded from optimization.

Implementation Environment

All training and evaluation was performed on a single workstation equipped with an NVIDIA GeForce RTX 4080 GPU (16 GB VRAM), a multi-core CPU, and 32 GB of system RAM. The implementation used Python 3.10, PyTorch 2.5.1 with CUDA 12.1, and Ultralytics 8.4.51 for the YOLO11 model definition and training loop. Federated simulation (B2) executed clients sequentially on this single GPU within each communication round, rather than concurrently across separate physical machines; this is consistent with the simulated, single-host framing of the federation described in the *Federated Partition* subsection above.

B1: Centralized Reference Baseline

B1 is a centralized reference baseline (not a mathematical upper bound), trained for a maximum of 150 epochs with an early-stopping patience of 30 epochs, batch size 16, and input size 960×960 . B1 is trained on the full training subset (8,937 images) in a single, non-federated run, using the same underlying training images that are subsequently partitioned across the four clients in B2; this ensures that any performance difference between B1 and B2 reflects the effect of federation and data partitioning rather than a difference in the underlying training data pool. The reported checkpoint is selected using

validation performance and subsequently evaluated once on the held-out test split.

B2: Non-IID Federated Training Protocol (FedAvg)

B2 applies FedAvg [12, 13] under the same $K=4$, Dirichlet $\alpha = 0.5$, partition-seed-42 configuration described above, using 40 communication rounds, 2 local epochs per round, SGD optimization (learning rate 0.01, momentum 0.9, weight decay 0.0005), batch size 8, input size 960, and no warmup. Training was repeated with three training seeds (42, 123, 2026, per the *Key Terminology* subsection above) that vary only the stochastic training realization and were not selected for any special numerical property; the partition seed remains fixed at 42 across all three runs. This replication is the basis for the reproducibility analysis reported in *Results* below.

Held-Out Test Protocol

The held-out test split is reserved strictly for final reporting and is not used to choose hyperparameters, communication rounds, or checkpoints. B1 was evaluated on the held-out test split only after its checkpoint was locked, and this result was not used to choose or modify the configuration of B2. For B2, each of the three locked seed checkpoints is intended for eventual evaluation on the same held-out test split; at the time of writing, B2 results are reported on the validation split, and validation and held-out test metrics are not mixed in direct comparisons within this manuscript.

Evaluation Metrics

Detection performance is reported using standard object-detection metrics: mAP50, mAP50-95, precision, and recall, computed on the split specified in each results table (the held-out test split for B1; the validation split for B2).

Reproducibility

Partition seed 42 is used for the federated partitioning underlying B2. B2 is repeated using three training seeds (42, 123, 2026) that vary only the stochastic training realization; the partition seed remains fixed across all three runs. Implementation code, dataset-splitting manifests, and configuration files are available from the corresponding author upon reasonable request.

RESULTS

This section reports results for the centralized reference baseline (B1) and the multi-seed Non-IID federated training protocol (B2), organized to separate outcome-level reproducibility (final performance across seeds) from trajectory-level reproducibility (convergence behavior across communication rounds).

Centralized Reference Performance

Table 3 reports the finalized held-out test results for B1.

Table 3: B1 centralized reference baseline — held-out test results

Class	AP50	AP50-95
Abnormal	0.9924	0.8462
Empty Bunch	0.9817	0.7521
Overripe	0.9406	0.6968
Ripe	0.5571	0.4757
Underripe	0.8275	0.7382
Unripe	0.9929	0.8766
Overall mAP	0.8820	0.7309

Overall Precision = 0.8745; Overall Recall = 0.8653. Ripe is currently the weakest-performing class in B1 by both AP50 and AP50-95. [TODO: Investigate and report the cause once confusion-matrix / dataset analysis is available; no cause is asserted here.] Consistent with the class structure described in the *Dataset* subsection above, the two structurally distinct classes (Abnormal, Empty Bunch) and the least visually ambiguous ripeness stage (Unripe) achieve the highest AP50 values (all ≥ 0.9817), while Ripe, a mid-sequence stage bordered by both Underripe and Overripe, achieves the lowest AP50 (0.5571). Underripe (0.8275), also a mid-sequence stage, shows an intermediate AP50 value broadly consistent with this pattern, though Overripe (0.9406) does not fit as closely; resolving this discrepancy is left to the confusion-matrix analysis noted above. Figure 4 presents the per-class AP50 values from Table 3 for visual comparison across the six ripeness-related classes, and Figure 5 provides qualitative detection examples, including at least one from the weaker-performing Ripe class.

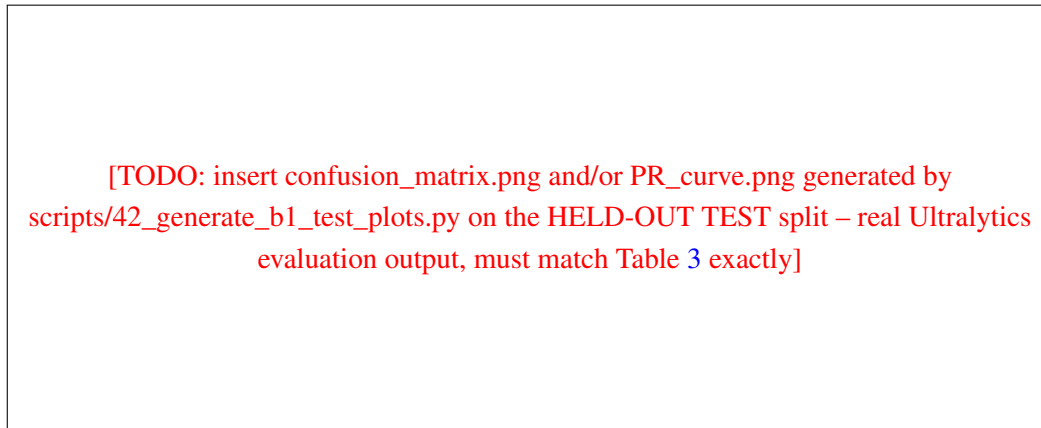


Figure 4: B1 (centralized) confusion matrix / precision-recall curve on the held-out test split.

[TODO: insert 2–4 held-out test images with model-predicted bounding boxes, class labels, and confidence scores overlaid (e.g., via Ultralytics’ built-in prediction visualization). Include at least one example from the Ripe class.]

Figure 5: Qualitative detection examples from B1 (and B2, where available), showing predicted bounding boxes, class labels, and confidence scores on held-out test images.

Multi-Seed Federated Validation Performance

Table 4 reports B2 validation results across the three training seeds. These are validation-set results; a direct B1-vs-B2 performance gap is not computed here, since B1 is reported on the held-out test split while B2 is reported on validation. A preliminary, explicitly non-matched comparison against B1 is given in *Matched Held-Out-Test Comparison (Preliminary)* below, and a fully matched held-out-test comparison is deferred to future reporting.

Table 4: B2 (FedAvg) — three-seed validation results

Seed	Best round	mAP50	mAP50-95	Prec. / Rec.
42	9	0.8850	0.7431	0.8717 / 0.8616
123	11	0.8594	0.7145	0.8139 / 0.8422
2026	40	0.8880	0.7544	0.8868 / 0.9104
Mean	—	0.8775	0.7374	0.8574 / 0.8714
SD	—	0.0157	0.0206	0.0385 / 0.0352

This outcome-level result – a small SD (0.0157) of validation mAP50 across three independent training seeds – is the primary reproducibility evidence reported in this study: it indicates that the federated configuration converges to a comparable operating point regardless of the stochastic training realization.

Convergence Trajectory Across Seeds

Figure 6 presents the per-round validation mAP50 trajectory for the three seeds and is the primary evidence for the trajectory-level component of this study’s reproducibility claim. The best-validation checkpoint round varied substantially across seeds: round 9 for seed 42, round 11 for seed 123, and round 40 (the final round) for seed 2026 – an approximately four-fold spread. This indicates that, although final validation performance is comparable across all three seeds (mAP50 range: 0.8594–0.8880), convergence *speed* under this Non-IID partition is sensitive to the stochastic training trajectory: outcome-level reproducibility does not imply trajectory-level reproducibility.

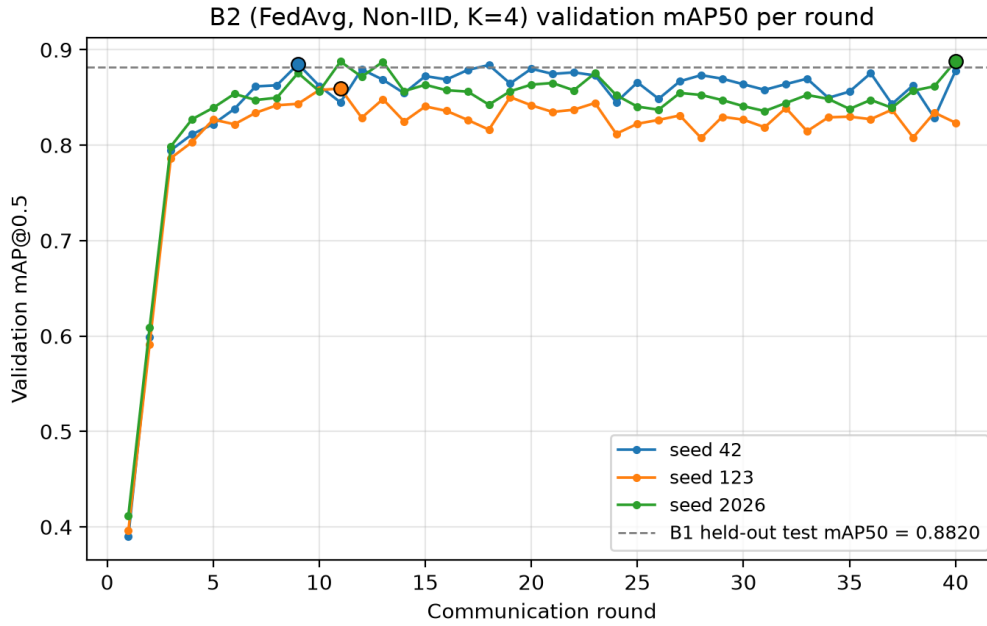


Figure 6: B2 (FedAvg, Non-IID, K=4) validation mAP50 per communication round across three independent training seeds (42, 123, 2026), with B1’s held-out test mAP50 shown as a reference line. Filled markers denote each seed’s best-validation round (Table 4), illustrating substantial trajectory-level variance (round 9–40) despite convergent final-outcome performance.

Best-Round versus Final-Round Behavior

Table 5 compares each seed’s best-validation checkpoint against its final-round (round 40) checkpoint. For seed 2026, the best and final rounds coincide by construction (best round = 40), so the difference is zero. [TODO: RESULT NOT YET AVAILABLE: round-40 validation mAP50 for seeds 42 and 123, needed to complete this table, pending re-evaluation of the corresponding saved checkpoints.]

Table 5: Best-validation versus final-round (round 40) checkpoint performance

Seed	Best round	Best mAP50	Final (round 40) mAP50	Δ (Best – Final)
42	9	0.8850	[TODO: TBD]	[TODO: TBD]
123	11	0.8594	[TODO: TBD]	[TODO: TBD]
2026	40	0.8880	0.8880	0.0000

Per-Class Federated Stability

Global mAP50 stability across seeds (Table 4) does not by itself establish that every individual ripeness class is equally stable; a class with high inter-seed variance could be masked by a low-variance aggregate metric if its AP50 magnitude is small relative to others. [TODO: RESULT NOT YET AVAILABLE: per-class AP50 for each of the three B2 seeds, needed to populate Table 6 below and directly test Supporting Research Question (iii).]

Table 6: Per-class AP50 across the three B2 training seeds (pending)

Class	Seed 42	Seed 123	Seed 2026	Mean	SD
Abnormal	[TODO: TBD]	[TODO: TBD]	[TODO: TBD]	—	—
Empty Bunch	[TODO: TBD]	[TODO: TBD]	[TODO: TBD]	—	—
Overripe	[TODO: TBD]	[TODO: TBD]	[TODO: TBD]	—	—
Ripe	[TODO: TBD]	[TODO: TBD]	[TODO: TBD]	—	—
Underripe	[TODO: TBD]	[TODO: TBD]	[TODO: TBD]	—	—
Unripe	[TODO: TBD]	[TODO: TBD]	[TODO: TBD]	—	—

Matched Held-Out-Test Comparison (Preliminary)

B1’s held-out test mAP50 (0.8820) and B2’s three-seed mean validation mAP50 (0.8775, SD = 0.0157) differ by 0.0045. This manuscript deliberately does not characterize this as a matched centralized-versus-federated comparison, since the two are evaluated on different splits (held-out test versus validation); the closeness is reported descriptively only, not as evidence of a specific, quantified federation cost. [TODO: RESULT NOT YET AVAILABLE: held-out test mAP50/mAP50-95 for each of B2’s three locked seed checkpoints, needed to complete Table 7 below and establish a genuinely matched comparison.]

Table 7: Matched held-out-test comparison (pending B2 held-out evaluation)

Model	Split	mAP50	mAP50-95
B1 (centralized)	Held-out test	0.8820	0.7309
B2 seed 42	Held-out test	[TODO: TBD]	[TODO: TBD]
B2 seed 123	Held-out test	[TODO: TBD]	[TODO: TBD]
B2 seed 2026	Held-out test	[TODO: TBD]	[TODO: TBD]

DISCUSSION

The multi-seed results reported here separate two distinct notions of federated-training reliability that are often conflated in single-run reports: *outcome-level reproducibility* (how similar the final achievable performance is across independent runs) and *trajectory-level reproducibility* (how similar the path – specifically, the number of communication rounds – taken to reach that performance is). B1’s held-out test results are reported here as a single reference point (mAP50 = 0.8820, mAP50-95 = 0.7309), with Ripe currently the weakest-performing class by a wide margin (AP50 = 0.5571 versus 0.8275 or higher for all other classes); a definitive cause cannot be asserted without confusion-matrix or dataset-level analysis, but plausible contributing factors include gradual visual transitions into and out of the Ripe stage and possible visual overlap with the adjacent Underripe and Overripe classes, which future work should investigate directly.

Outcome-Level Reproducibility

The three B2 seeds converge to a comparable peak validation mAP50 (0.8594–0.8880, SD = 0.0157), which is consistent with – though not proof of – FedAvg being a numerically stable aggregation procedure under this specific Non-IID partition (Dirichlet $\alpha = 0.5$, $K=4$). This is the strongest reproducibility evidence in this study, since it is directly measured across three independent stochastic training realiza-

tions rather than inferred from a single run.

Trajectory-Level Reproducibility

The same three seeds reach their best checkpoint at markedly different rounds (9, 11, and 40), an approximately four-fold spread. This is an interpretation, not a proven causal claim: a single-seed federated experiment reporting only “round 9, mAP50 = 0.8850” would have substantially understated the number of rounds this configuration can require to converge, illustrating why single-seed federated reports risk incomplete conclusions about training-time budgets. Distinguishing outcome- from trajectory-level reproducibility is, to the authors’ knowledge, not commonly reported jointly in federated object-detection studies (see *Related Work*), and is the central methodological point of this manuscript.

Client Heterogeneity and Convergence

Client 2 contributes 59.36% of the training subset (Table 2), substantially more than the remaining three clients combined ($CV \approx 0.836$; *Client Heterogeneity Measurement*, Materials and Methods). Because FedAvg aggregates client updates weighted by local sample count [12], Client 2’s locally well-supported gradient signal plausibly exerts disproportionate influence on the aggregated global update each round; this is offered as a plausible contributing hypothesis for the observed outcome-level stability, not as a demonstrated causal mechanism – confirming it would require a controlled ablation (e.g., uniform-weighted aggregation) that is outside the scope of the present study. This quantity imbalance is distinct from, and should not be conflated with, class-distribution imbalance across clients, which this manuscript does not currently characterize (*Client Heterogeneity Measurement*).

Per-Class Stability and the Preliminary Centralized–Federated Comparison

Per-class stability across seeds has not yet been evaluated in this manuscript (*Per-Class Federated Stability*, Results); global mAP50 stability does not guarantee that every class is individually stable. Similarly, the numerical closeness between B1’s held-out test mAP50 (0.8820) and B2’s validation mean (0.8775) is reported descriptively only (*Matched Held-Out-Test Comparison (Preliminary)*, Results) and should not be read as evidence of a specific, quantified federation cost, since establishing that requires evaluating the locked B2 checkpoints on the same held-out test split used for B1.

Scope of the Heterogeneity Studied

This study varies only *statistical* client heterogeneity (via Dirichlet partitioning) under a single-host sequential simulation; it does not exercise distributed-systems behavior such as client dropout, communication failure, or network latency, which are separate and equally relevant dimensions of real-world federated deployment reliability.

Future Work: Differential Privacy

A planned extension of this pipeline will incorporate sample-level DP-SGD [19] to quantify the additional utility cost of formal per-sample privacy guarantees, building on the reproducibility baseline established here. Results for this extension are outside the scope of the present study.

Limitations

Federated training was simulated sequentially on a single GPU (single-host simulation); no distributed-systems factors (network latency, client dropout, communication failure) were exercised. The dataset

originates from a single public source, so generalization to other plantations, cultivars, cameras, or lighting conditions is not directly evaluated. The four clients are simulated sample-level data shards drawn from one training subset, not four physically distinct plantations, estates, organizations, or devices; conclusions are limited to the statistical heterogeneity captured by the Dirichlet partition. K is fixed at 4 and α is fixed at 0.5; sensitivity of the reported reproducibility findings to other client counts or Non-IID severities has not been tested. Only three training seeds were used, which establishes a first-order variance estimate but a larger number of replications would tighten it. Client heterogeneity is currently characterized only by training-image count, not by class distribution or bounding-box count. B2 has not yet been evaluated on the held-out test split, so no matched centralized-versus-federated comparison is currently possible. No BatchNorm-versus-GroupNorm ablation has been run, so Group Normalization's effect on the observed stability is not empirically isolated from other factors. This study does not provide a formal (Differentially Private) privacy guarantee; the federated setup here only avoids transmitting raw images, which is a narrower property than formal privacy.

CONCLUSION

This study evaluated the reproducibility and convergence stability of Non-IID federated training (FedAvg, Dirichlet $\alpha = 0.5$, $K=4$) for six-class oil palm FFB ripeness detection using YOLO11n with Group Normalization, replicated across three independent training seeds over 40 communication rounds each. Outcome-level reproducibility was strong: validation mAP50 converged to a stable mean of 0.8775 (SD = 0.0157) across seeds. Trajectory-level reproducibility was substantially weaker: the best-validation checkpoint occurred at round 9, 11, and 40 for the three seeds respectively, a roughly four-fold spread, despite substantial client data imbalance (8.67% to 59.36% of training images per client). This distinction – a stable final outcome reached via seed-dependent convergence paths – is the central finding of this study and motivates multi-seed reporting as a necessary, not merely desirable, practice for characterizing federated training behavior under Non-IID partitions. A centralized reference model, evaluated once on a leakage-free held-out test split, achieved mAP50 = 0.8820; this is reported descriptively alongside the federated results, and a fully matched held-out-test evaluation of the federated checkpoints remains the immediate next step before any quantified centralized-versus-federated utility gap can be claimed. These findings establish a reproducibility baseline for a planned future extension of this pipeline incorporating Differentially Private SGD.

CONFLICT OF INTEREST

The authors declare that there is no conflict of interest between the authors or with the research object of this paper.

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DECLARATION OF GENERATIVE AI USE

During the preparation of this manuscript, the author(s) used Claude (Anthropic), a large language model-based AI assistant, to help draft, restructure, and format portions of the manuscript text and to

prepare the accompanying LaTeX and Word document files, based on experimental details, results, and constraints supplied directly by the author(s). The AI tool did not generate, fabricate, or interpret any experimental results. All AI-assisted content was reviewed, verified, and edited by the author(s), who take full responsibility for the accuracy, originality, and integrity of the published work.

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